አብነታት ልሳን ዐቢይ

Large language models

Weekend 1: Language models with the Ge'ez games

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 16 October 2026 ፳፻፳፮

We start our first module with an understanding of large language models

ግዕዝ

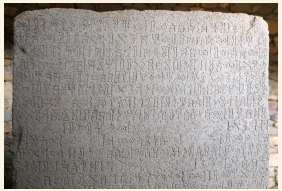
Introduction

Ge’ez

To make things more vivid, I’m going to use the language “Ge’ez” as our illustrative example

ግእዝ ቋንቋ::

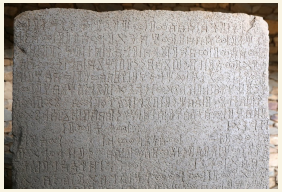
- ጥንታዊ ደቡባዊ ሴማዊ ቋንቋ:: ይህ ቋንቋ ከአቢሲኒያ የመጣ ነው፤ አቢሲኒያ ዛሬ ኢትዮጵያና ኤርትራ ተብሎ ታወቃለች::
- ይህን ቋንቋ መሠረት አድርገን ዘመናዊ ትልቅ ቋንቋ ሞዴሎችን እመማርና እምርምር እንጠቀማለን::



giving us a window into the history of the region

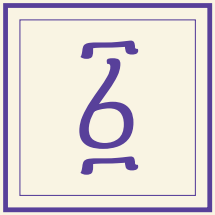
The Ge’ez language

- Ancient South Semitic language. The language originates from Abyssinia, what is now Ethiopia and Eritrea.
- We use this language as basis to study and investigate modern large language models.



The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.



Section 1

Structure

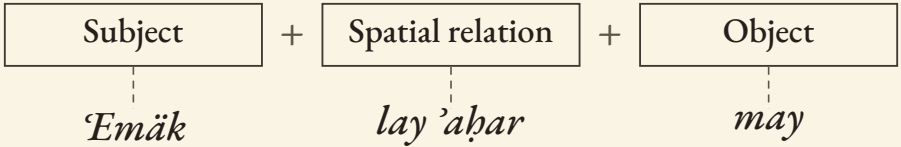
Game 1: objective

- We will show you some sentences in Ge'ez. They are related to either **city entities (city)** or **rural entities (countryside)**.
- Your task is to learn to discover when a sentence is talking about something in the city and when it is talking about something in the countryside.

Countryside	City
ʿEmäk lay ʾaḥar may	*bet nawīd lay ʾaḥar ʿIqlēsīya*
gāzā lay ʾasmāl may	*maytān lay ʾasmāl ʿIqlēsīya*
may lay ʾaḥar gāzā	*ʿIqlēsīya lay ʾaḥar maytān*
ṣadäq lay ʾaḥar ʿEmäk	*bet masḥāf lay ʾaḥar bet nawīd*
ṣadäq lay ʾaḥar gāzā	*bet masḥāf lay ʾaḥar maytān*

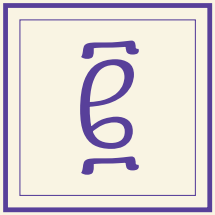
Insight

- We do not need to understand the meaning of a language nor its components to be able to extract information from it.
- We are dealing with a very simple language:



The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.



Section 2

Imitation

Game 2: objective

- After observing sufficiently many examples of Ge'ez sentences you are now capable of producing not only syntactically valid, but also semantically meaningful sentences! Give it a try.
- For example:
  - "Carlos is at ETH" is **semantically valid!**
  - "ETH is at Carlos" is **not semantically valid!**

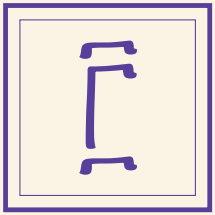
Insight

We don't need to understand a language in order to produce meaningful sentences!



The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.



Section 3

Prediction

Prediction

	ዕድል Probability	Language models can be described as **probabilistic trees**.
	ሰንጠረዥ Table	**Probabilistic trees** can be represented with **tables**.
	ሥልጠና Learning	**Tables** can be learned from language texts.
	የውይይት ሮቦት Chatbot	The **table** can be used as a **chatbot**.

Game 3: Prediction

Alice plays tennis

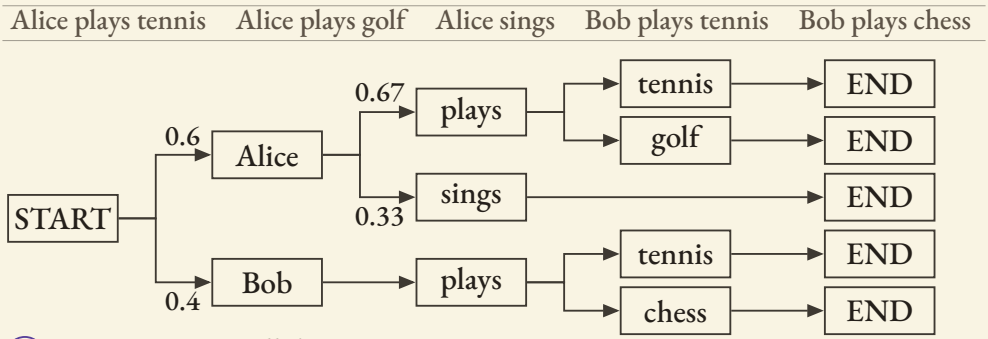
Alice plays golf

Alice sings

Bob plays tennis

Bob plays chess

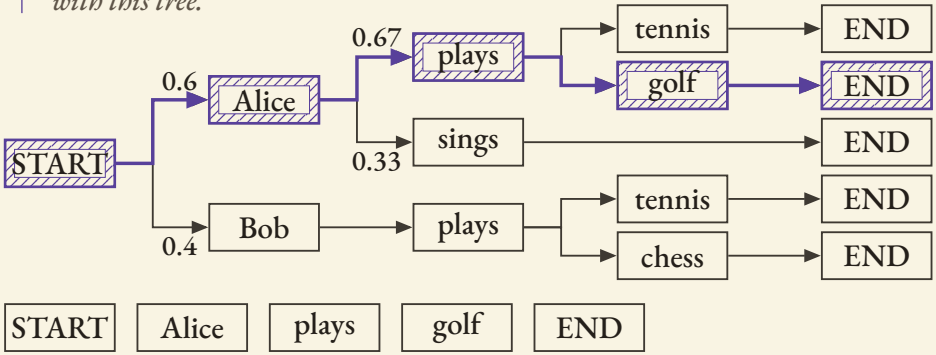
Probability trees



We can organize all these sentences into a probability tree!

Text generation as sampling

We can now produce sentences with this tree.



Insight: We can represent a language model with a table!

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0



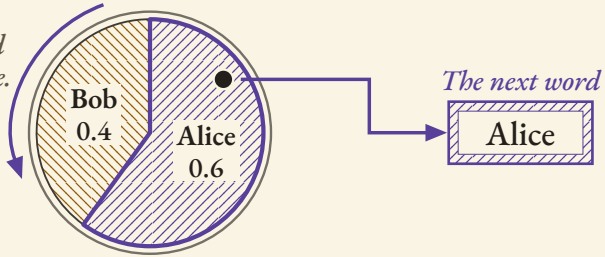
A blank cell is a chance of 0: no sentence of the corpus goes that way.

Sampling: spin a roulette

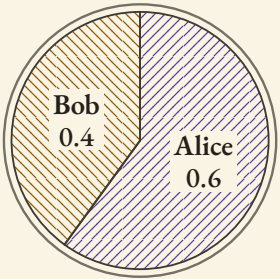
Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						

Each slice is as big as the chance of its word. A word with no chance gets no slice.

Spin the roulette.



A bigger slice comes out more often



Ten spins, on average

Alice

Bob

A real run of ten spins can differ a little. The more spins, the closer the shares get to 6 in 10 and 4 in 10.

We can now generate sentences by iterative sampling.

START Alice plays golf END

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

Training an autoregressive language model

Alice likes fruits

Alice likes candies

Bob likes fruits

(START), Alice
(START), Alice
(START), Bob
(Alice), likes
(Alice), likes

(Bob), likes
(Alice, likes), fruits
(Alice, likes), candies
(Bob, likes), fruits
(Alice, likes, fruits), END
...

Prefix	Alice	Bob	likes	candies	fruits	END
START	2/3	1/3				
Alice			1			
Bob			1			
Alice likes				1/2	1/2	
Bob likes					1	
...						

Your first chatbot!

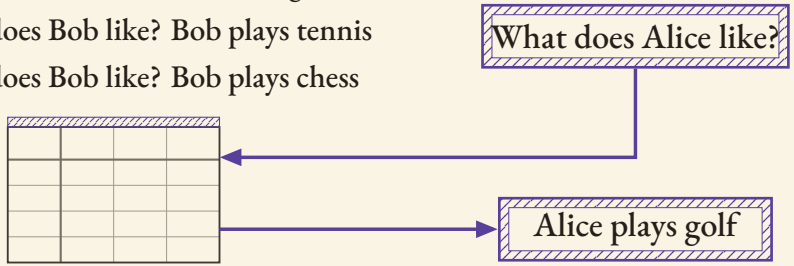
What does Alice like? Alice plays tennis

What does Alice like? Alice plays golf

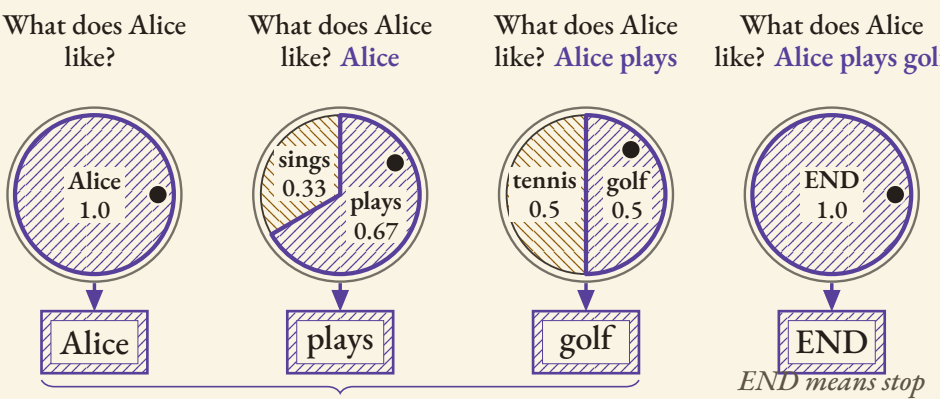
What does Alice like? Alice sings

What does Bob like? Bob plays tennis

What does Bob like? Bob plays chess



A chatbot answers one word at a time



Summary

- Language models can be described as probabilistic trees.
- A language model can be represented computationally as a **table** that describes all relevant **conditional probabilities**.
- In our toy, a **table** can be learned from a set of training sentences, by generating all combinations of prefix and continuation, and then calculating probabilities in the corresponding table.
- In our toy, a conversation can then be carried out, by just taking the user's prompt, and using the **table** to generate **probable continuations**.

The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.